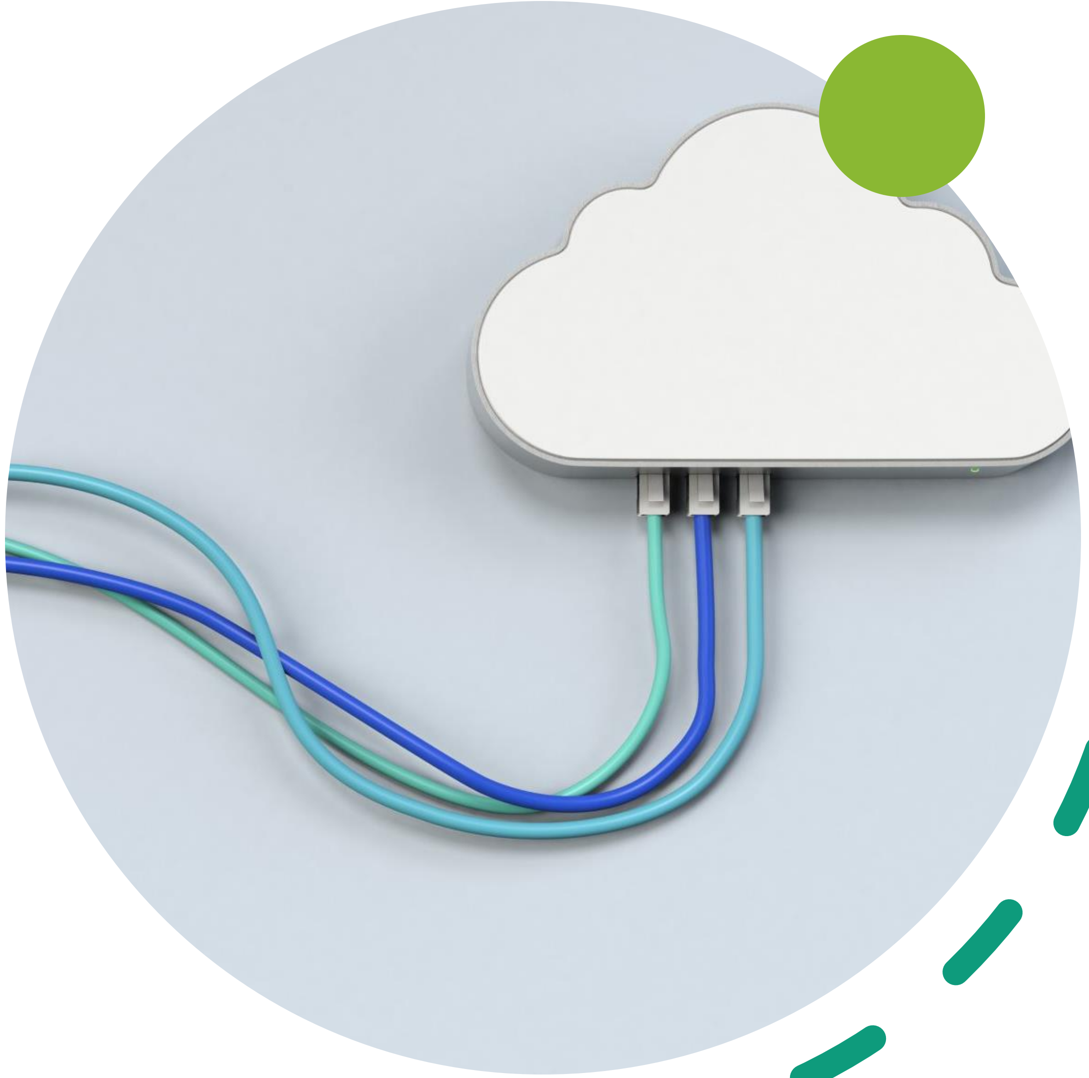
Cloud Security Leveraging AI: A Fusion-Based AISOC for Malware and Log Behaviour Detection

- Name: Nnamdi Philip Okonkwo
- Program: MEng Cybersecurity, University of Limerick
- Student ID: 24013366
- Supervisor: Dr. Lubna Luxmi Dhirani

27-08-2025



Problem Statement

High-volume, heterogeneous telemetry from elastic cloud resources

Stealthy, adaptive attacks (malware & log-based) designed to evade conventional systems.

Need for cost-efficient, real-time threat identification in cloud Security Operations Centers (SOCs).

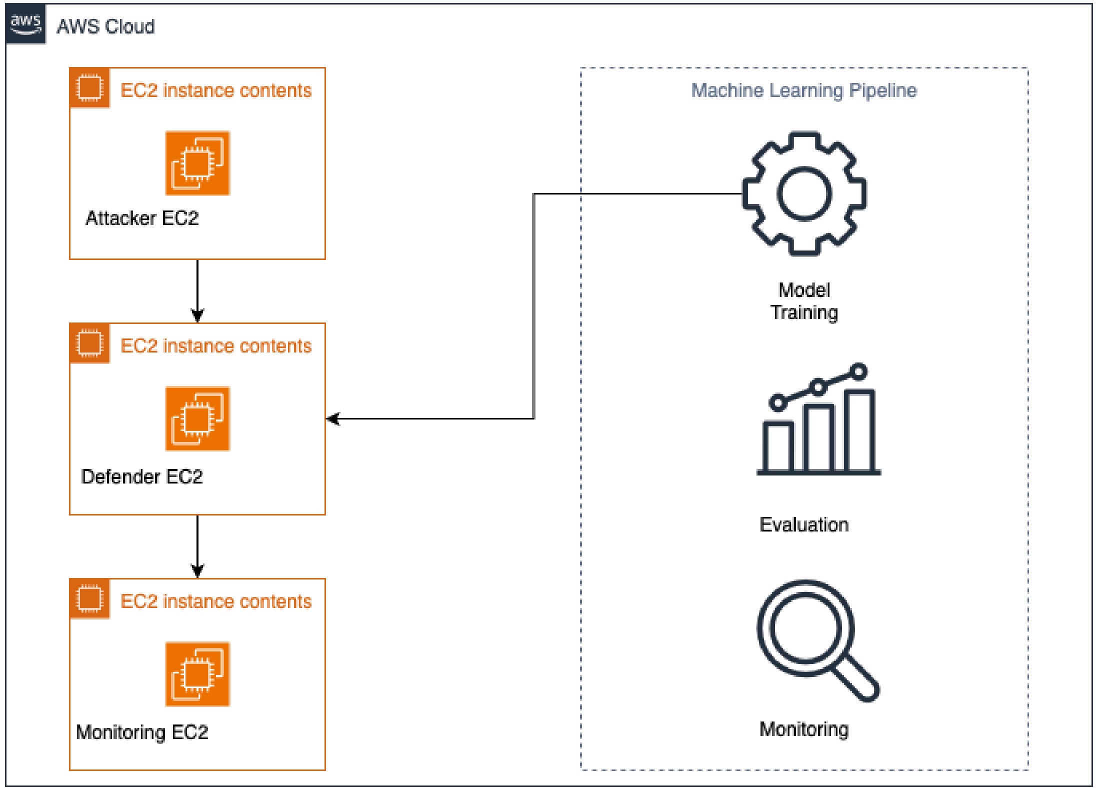
Our Solution: AI-Augemented SOC (AISOC) on AWS

Simulated Cloud Environment: Three AWS EC2 instances (Attacker, Defender, Monitoring).

Attack Simulation: Metasploit reverse-shell from Attacker to Defender, with Filebeat forwarding logs to ELK stack (Elasticsearch/Kibana).

Multi-Modal Fusion: Combining model outputs into a three-level alert taxonomy.

Dual-Model Detection: AI models for malware & log anomaly detection.



Key AI Models and Techniques

Malware Detector: Random Forest Classifier trained on UCI Malware Detection Dataset

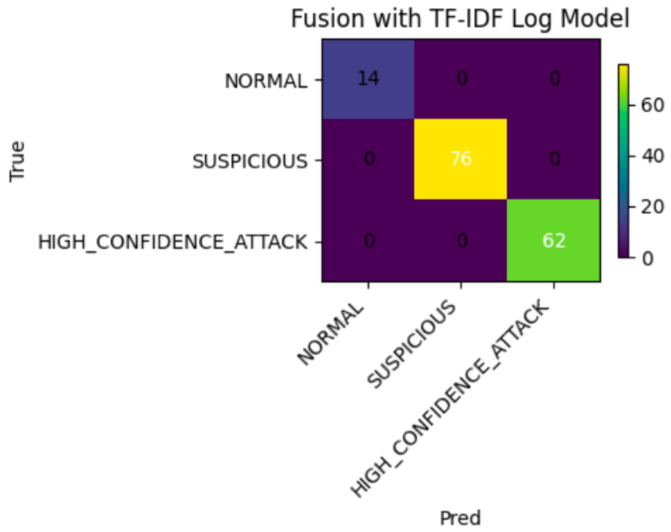
Log Anomaly Detector: Logistic Regression with TF-IDF features.

- Adversarial Log Augmentation: Keyword obfuscation, noise injection, random character swap, synonym replacement to improve robustness against evasion.

Fusion Logic and Thresholds

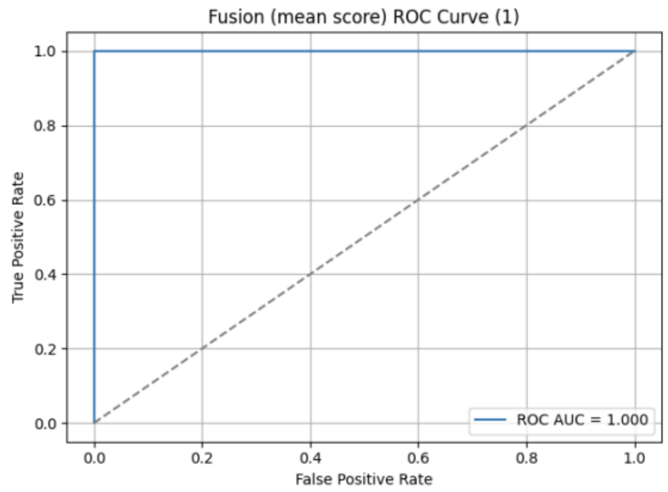
Alert Levels: NORMAL, SUSPICIOUS (one modality signals), HIGH_CONFIDENCE_ATTACK (both modalities exceed thresholds).

Tuned Thresholds: Malware score ≥ 0.10, Log score ≥ 0.42 (maximized macro F1 on validation set).



Performance Highlights

- Malware Model: 1.00 Precision, Recall, F1-score on validation and test sets
- Augmented Log Model: Cross-validation macro F1-score of 0.91 (after augmentation).
- Final Fusion System: Macro F1-score of 1.00 across all alert levels on the held-out test set.
- ROC/PR Curves: Confirmed robustness and reliability.



Conclusion

Caveats

- Idealized Evaluation Conditions
- Operational Metrics Not Empirically Measured
- Limited Adversarial Robustness (Adversarial Training or Shift Detection is needed)
- Narrow Telemetry Scope
- Batch Processing

Future Work

- Implementing real-time monitoring using cloud schedulers
- Incorporate more diverse telemetry
- Developing smarter attack simulations using techniques like reinforcement learning to further harden the models against evasion.